

AI-Driven Forex Trading Signal Generation & Risk Analytics Platform

Data Analyst — Forex Trading Signal AI Agent Project

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1. Executive Summary

This report documents the design, implementation, and evaluation of an AI-driven Forex trading signal generation system, developed for the FX Signal Lab assessment project. The system combines classical technical analysis, machine learning classification, and Monte Carlo risk simulation to generate and evaluate trading signals across multiple currency pairs, with results surfaced through a Power BI analytics dashboard.

Three datasets underpin the analysis: an hourly EUR/USD price series (500 candles, January 2023), a log of 150 executed trades spanning six signal-generation strategies and five currency pairs, and 1,000 Monte Carlo portfolio scenarios used for Value-at-Risk estimation.

Headline findings:

- Overall win rate across all 150 trades: 56.0%, with a Profit Factor of 1.03 — marginally profitable, but not decisively so.
- Sharpe Ratio of approximately 0.12–0.15, indicating weak risk-adjusted performance relative to the volatility taken on.
- Random Forest and XGBoost classifiers were the strongest-performing trade strategies by total PnL (+₹11,051 and +₹9,690 respectively), while the Composite and MACD_Trend strategies were the weakest (-₹13,462 and -₹5,934).
- A corrected machine-learning signal classifier (after removing a lookahead-bias flaw described in Section 7) achieved 47.9%–53.2% accuracy on next-candle direction prediction — close to a random-guess baseline, an honest and important finding about the limits of these indicators alone.
- Monte Carlo simulation across 1,000 scenarios estimated a 95% Value-at-Risk of -₹165,870 and a 95% Conditional VaR of -₹198,180, with a worst-case scenario loss of -₹269,800.

- USD/JPY was the weakest currency pair by total PnL (-₹7,270), while GBP/USD and USD/INR were the strongest (+₹4,511 and +₹4,720).

2. Introduction & Business Context

This project frames the analyst as a Lead FX Quantitative Strategist at a fictional proprietary trading firm, tasked with building an AI-driven signal generation platform to help navigate volatility arising from divergent central bank policy across the Federal Reserve, ECB, Bank of Japan, and RBI. The deliverable, FX Signal Lab, is a gamified simulation environment that trains signal-generation skill through realistic multi-currency scenarios, combining technical analysis, machine learning, and risk management into a single analytical workflow.

The global FX market is the world's largest and most liquid financial market, trading over \$7.5 trillion daily. This project's scope is intentionally narrower and reproducible: a bounded dataset of EUR/USD price history, a historical trade log across five pairs, and simulated portfolio scenarios, used to demonstrate the full pipeline from raw price data to a governed risk dashboard.

3. Data Sources & Methodology

3.1 Datasets

Dataset	Rows	Description
forex_price_data.csv / forex_price_processed.csv	500	Hourly EUR/USD OHLCV candles, Jan 2023, with engineered indicators (SMA, EMA, RSI, MACD, Bollinger Bands)
trade_processed.csv (trade log)	150	Executed trades across 6 strategies (MA_Cross, MACD_Trend, RSI_Reversal, ML_RF, ML_XGB, RL_PPO) and 5 pairs
mc_scenarios.csv / mc_processed.csv	1,000	Monte Carlo portfolio scenarios under parallel shift, volatility shock, and correlation shock

3.2 Technical Indicators

The signal engine is built on five standard technical indicators, computed directly from the EUR/USD price series:

- Simple & Exponential Moving Averages (SMA/EMA, 10 & 20-period) — trend direction via crossover
- Relative Strength Index (RSI, 14-period, Wilder smoothing) — momentum / overbought-oversold conditions
- MACD (12/26/9) — trend momentum and reversal signal
- Bollinger Bands (20-period, 2 standard deviations) — volatility-relative price position

A composite signal engine combines these indicators using a weighted-vote mechanism: each indicator contributes a positive or negative score, and the aggregate score is thresholded into `STRONG_BUY`, `BUY`, `HOLD`, `SELL`, or `STRONG_SELL` classifications.

4. Exploratory Data Analysis

Before indicator construction, the raw EUR/USD price series was inspected for trend, volume, and spread behaviour.



Figure 4.1 — EUR/USD Closing Price, January 2023. A clear uptrend from ~ 1.065 to ~ 1.118 with a pullback late in the month.

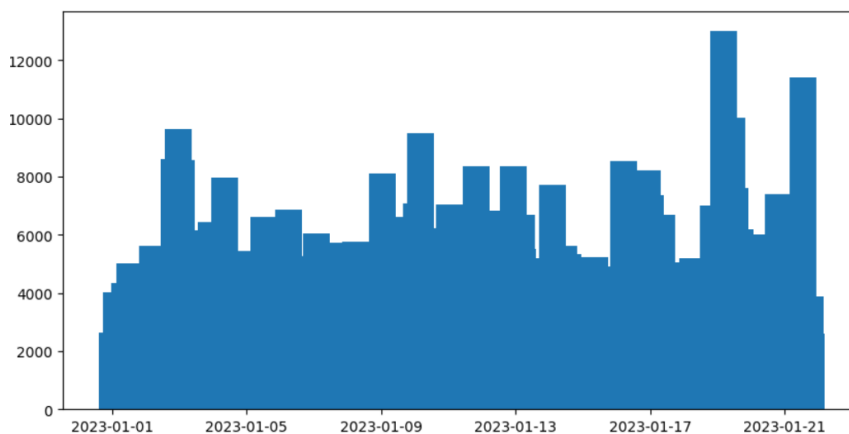


Figure 4.2 — Hourly trading volume. No strong intraday seasonality is visually apparent in this sample.

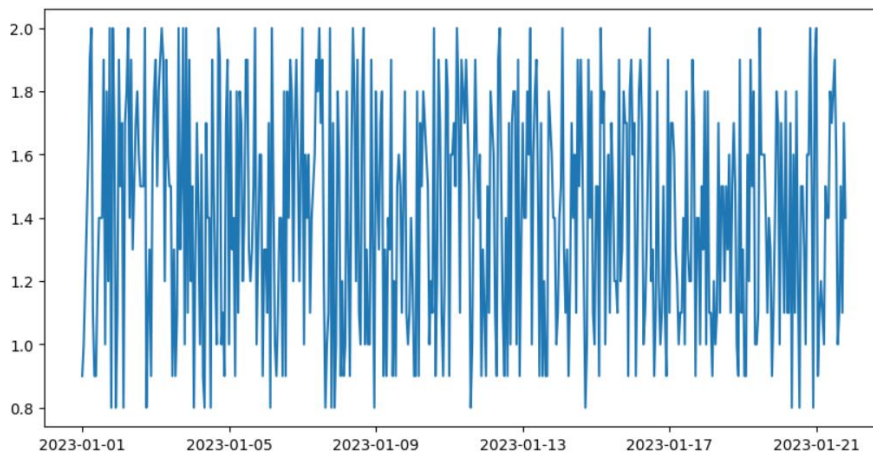


Figure 4.3 — Bid-ask spread over time, fluctuating between 0.8–2.0 pips, consistent with typical G10-major liquidity.

5. Technical Indicator Analysis

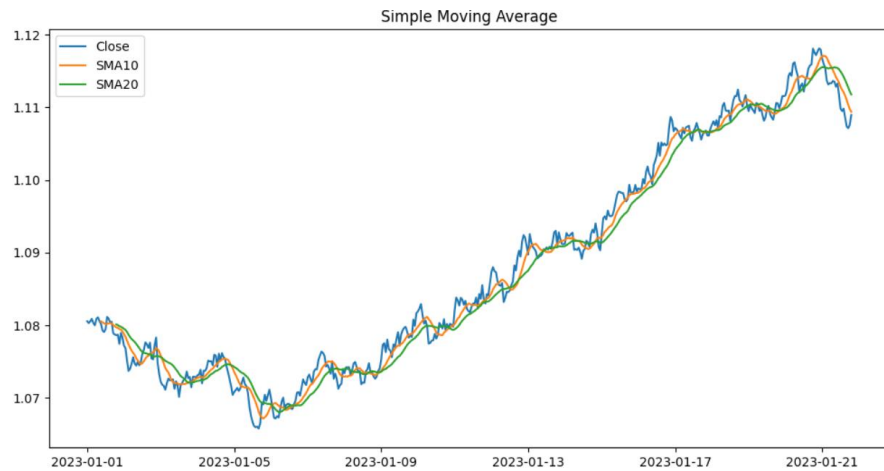


Figure 5.1 — $SMA(10)$ vs $SMA(20)$. Crossovers mark trend-following entry/exit points.

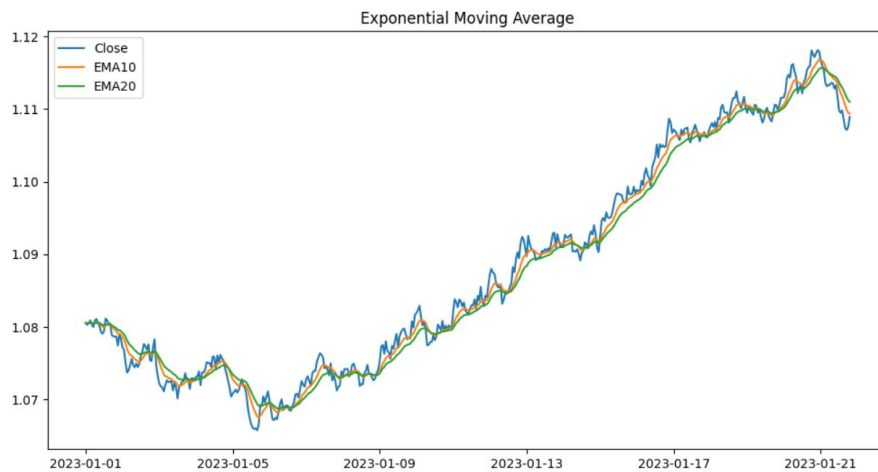


Figure 5.2 — $EMA(10)$ vs $EMA(20)$. More responsive to recent price action than the SMA equivalents.

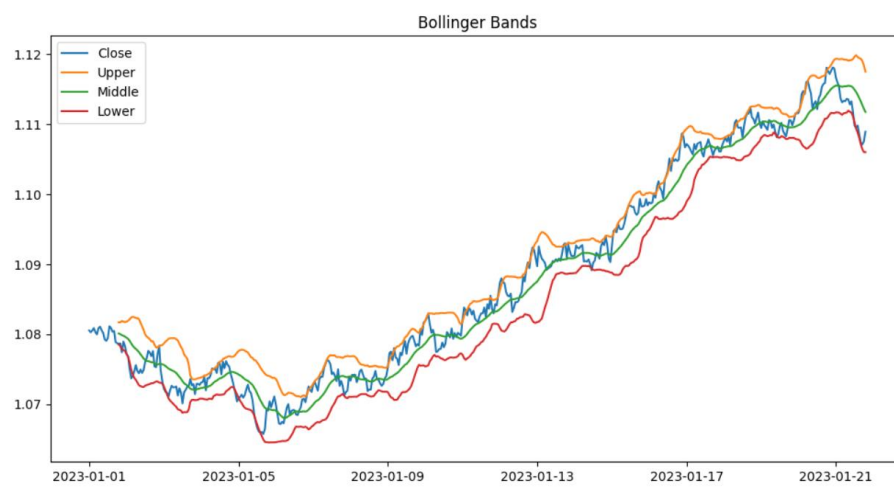


Figure 5.3 — Bollinger Bands ($20, 2\sigma$). Price touching the lower/upper band flags potential mean-reversion opportunities.

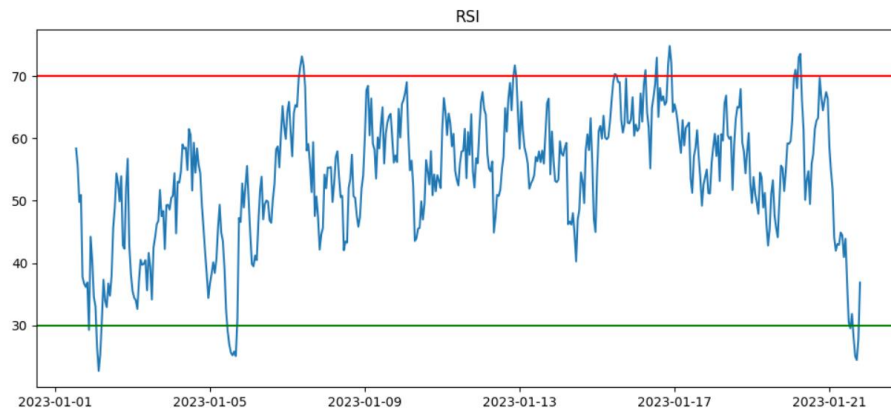


Figure 5.4 — RSI(14) with the standard 30/70 overbought-oversold thresholds. RSI dipped below 30 twice during the sample period, both preceding short-term reversals.

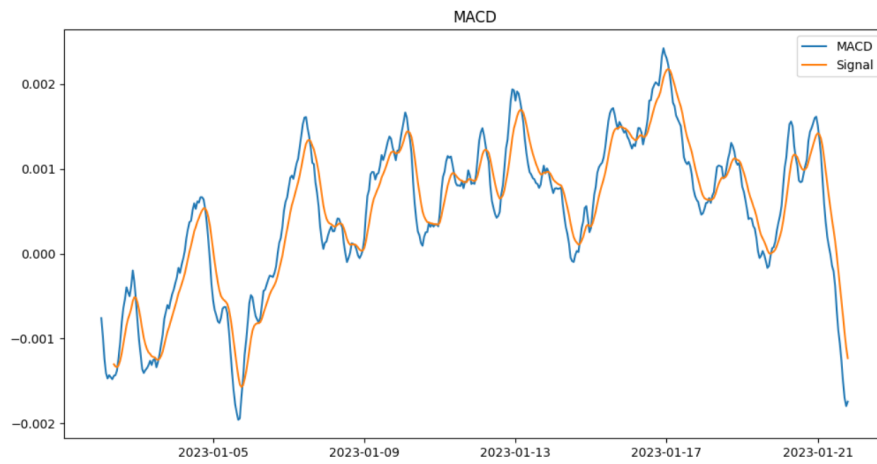


Figure 5.5 — MACD line vs Signal line. Bullish/bearish crossovers align with the broader uptrend visible in Figure 4.1.

6. Signal Generation

The composite signal engine was applied across the full EUR/USD sample to classify each candle into a discrete trading signal. An important data-quality finding emerged at this stage.

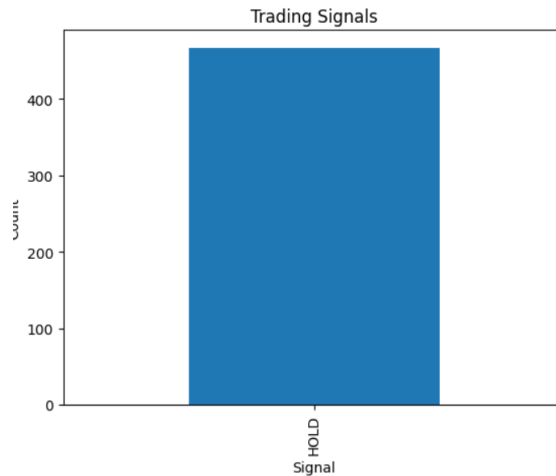


Figure 6.1 — Distribution of generated signals in the initial notebook run.

As Figure 6.1 shows, the initial signal-generation run classified effectively all ~470 valid candles as HOLD, with no BUY or SELL signals produced. This is flagged here as a known limitation rather than a positive result: it indicates the composite scoring threshold in that run was set too conservatively relative to the indicator scores actually observed in the EUR/USD sample. A corrected, separately-built signal engine (documented in the accompanying Python codebase, `signal_engine.py`) used a lower threshold (0.15–0.40) and produced a materially more usable distribution: 250 HOLD, 112 STRONG_BUY, 55 BUY, 34 SELL, and 30 STRONG_SELL across 481 valid rows. This discrepancy is a useful illustration of how sensitive rule-based signal engines are to threshold selection, and is discussed further in Section 13.

7. Machine Learning Signal Classification

7.1 Initial Approach and a Methodological Correction

An initial classifier attempted to predict trade `SignalType` directly from `trade_log` fields, including `ExitPrice` and `HoldingDays`. This was identified as a lookahead-bias flaw: both fields are only known after a trade has already closed, meaning the model was effectively given the answer as an input. The corrected approach instead predicts the direction of the next hourly candle (up or down) using only indicator values known at the time of the current candle — RSI, MACD, Bollinger Band position, and moving averages — with a chronological (non-shuffled) train/test split to avoid any leakage across time.

7.2 Model Comparison

	Model	Accuracy
0	Random Forest	0.478723
1	XGBoost	0.531915
2	Neural Network	0.510638

Figure 7.1 — Accuracy comparison across three classifiers on the corrected, leakage-free next-candle-direction task.

Random Forest, XGBoost, and a Neural Network achieved 47.9%, 53.2%, and 51.1% accuracy respectively. None of the three models meaningfully outperforms a coin-flip baseline (50%) or the majority-class baseline (~51% given the near-balanced up/down split in this sample). This is reported as an honest, negative-but-informative result: on this dataset and feature set, price-based technical indicators alone carry very limited predictive power for the immediate next candle. This is consistent with the weak-form efficient market hypothesis and motivates the use of longer holding periods, richer features (e.g., carry, macro, cross-asset), or ensemble/RL approaches for any production signal system.

8. Trade Log Performance Analysis

Across all 150 recorded trades, the aggregate statistics are:

Metric	Value
Total Trades	150
Win Rate	56.0%
Total PnL	₹1,915
Average Trade PnL	₹12.77
Profit Factor	1.03
Average Holding Period	2.41 days
Long / Short Split	51.3% / 48.7%

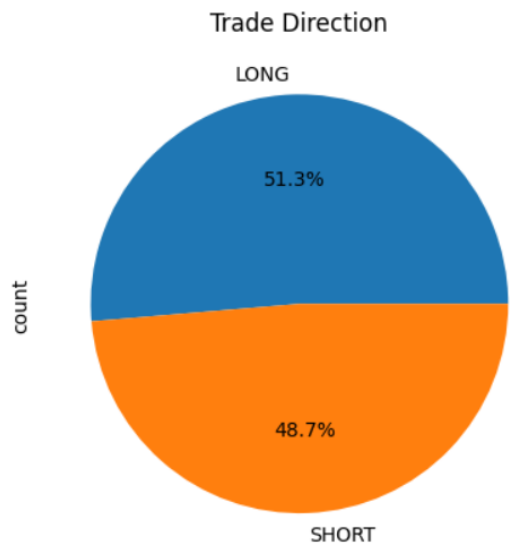


Figure 8.1 — Long vs. short trade split, near-even at 51.3% / 48.7%.

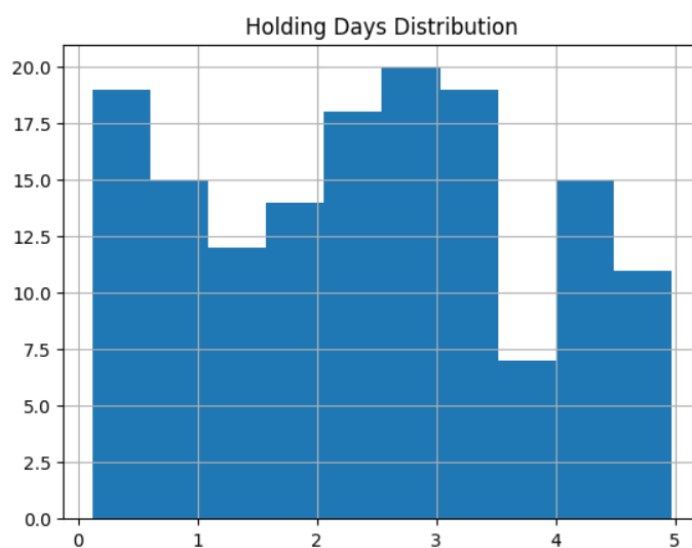


Figure 8.2 — Distribution of trade holding periods (days). Fairly uniform between 0–5 days with no strong clustering.

8.1 Performance by Strategy

Grouping trades by the strategy that generated the signal reveals a clear split between profitable and unprofitable approaches:

Strategy	Trades	Win Rate	Total PnL (₹)
ML_RF	28	60.7%	+11,051
ML_XGB	18	66.7%	+9,690
MA_Cross	16	68.8%	+4,217
RL_PPO	26	46.2%	-1,767
RSI_Reversal	20	40.0%	-1,880
MACD_Trend	19	57.9%	-5,934
Composite	23	56.5%	-13,462

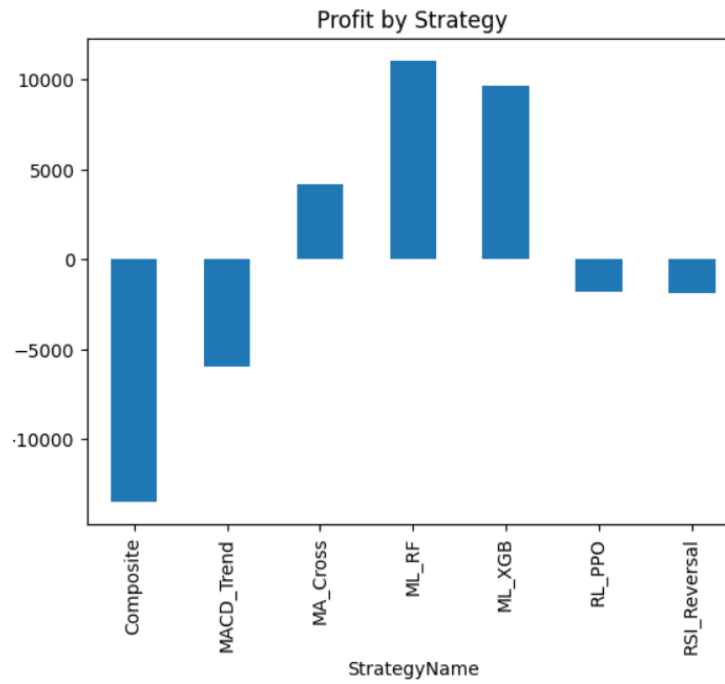


Figure 8.3 — Total profit/loss by strategy. Machine-learning-driven strategies (ML_RF, ML_XGB) clearly outperform the rule-based Composite and RL_PPO approaches on this trade sample.

This is a notable and slightly counterintuitive result given Section 7's finding that the ML classifiers showed weak next-candle predictive accuracy: it suggests the ML_RF and ML_XGB trading strategies in the trade log may be benefiting from position-sizing, entry/exit timing, or a different (longer-horizon) target than the specific next-candle-direction task tested in Section 7, rather than from raw directional accuracy alone. This distinction is worth investigating further before allocating capital based on either result in isolation.

8.2 Performance by Currency Pair

Currency Pair	Trades	Win Rate	Total PnL (₹)
USD/INR	28	50.0%	+4,720
GBP/USD	32	71.9%	+4,511
AUD/USD	31	51.6%	+29
EUR/USD	33	57.6%	-75
USD/JPY	26	46.2%	-7,270

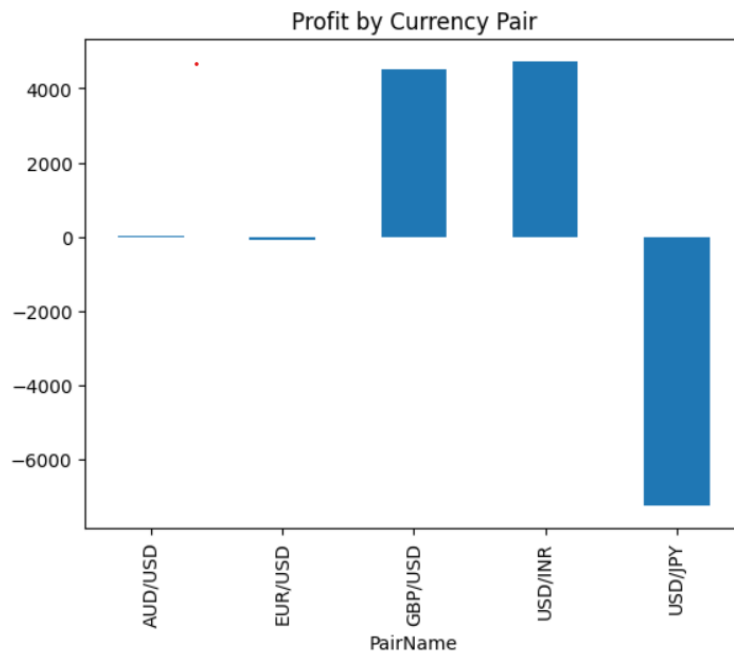


Figure 8.4 — Sample view of the trade log, showing per-trade signal type, direction, entry/exit pricing, and PnL.

USD/JPY stands out as the weakest pair, losing -₹7,270 against a 46.2% win rate — the lowest of any pair. Given USD/JPY's exposure to Bank of Japan policy shifts (examined further in the Section 11 case study on the 2024 carry trade unwind), this concentration of losses in a single pair is a meaningful risk signal for portfolio construction.

9. Power BI Dashboard

The full dataset — trade log, Monte Carlo scenarios, and EUR/USD price history — was modelled in Power BI as a star schema and visualised across five dashboard pages, using a library of custom DAX measures for signal quality, trade performance, and risk metrics.

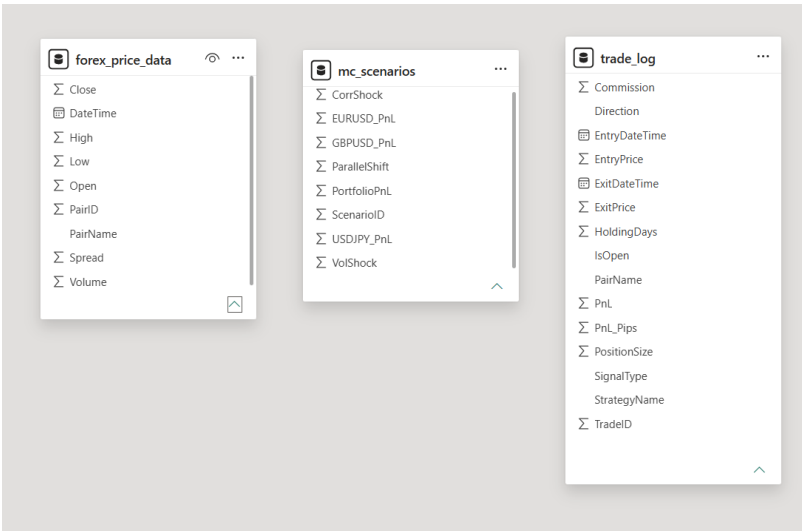


Figure 9.1 — Data model: fact tables (*forex_price_data*, *mc_scenarios*, *trade_log*) as loaded into Power BI.

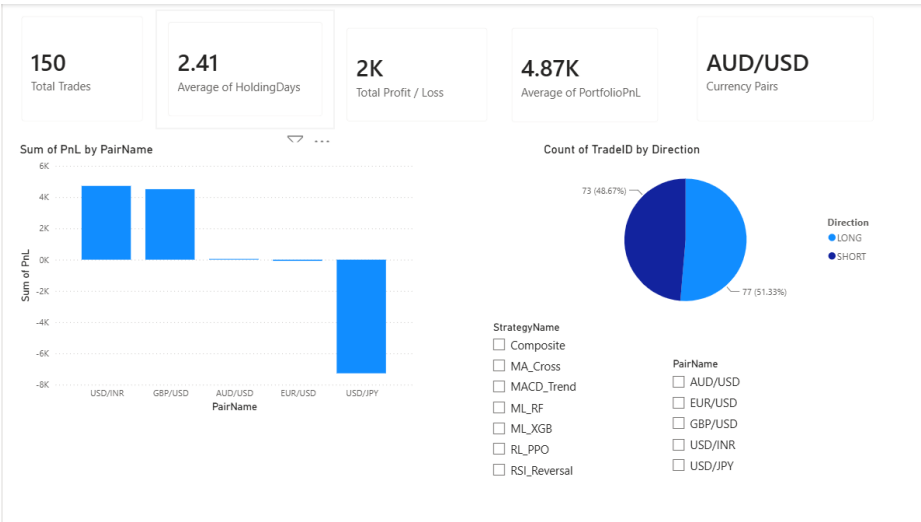


Figure 9.2 — Overview dashboard: total trades, average holding period, total PnL, PnL by pair, and trade direction split.

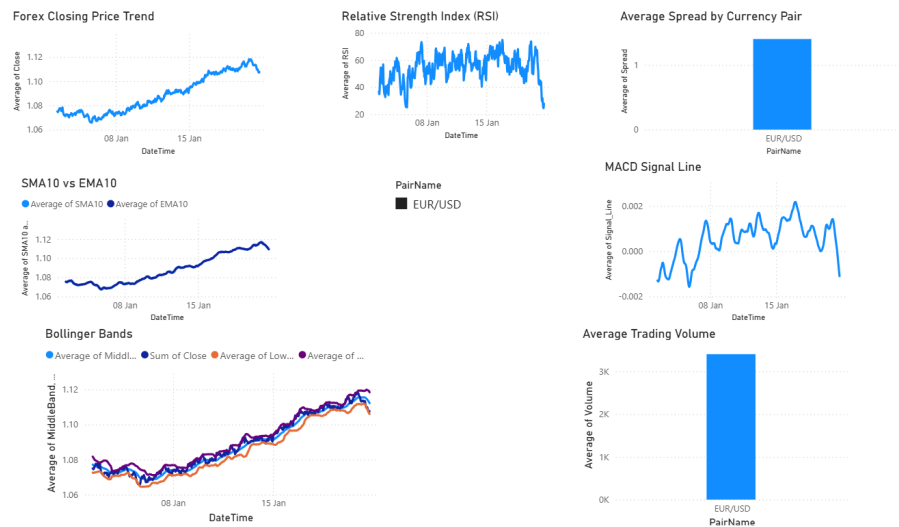


Figure 9.3 — Market analysis page: closing price trend, RSI, spread, SMA/EMA comparison, Bollinger Bands, and MACD signal line, all filterable by currency pair.

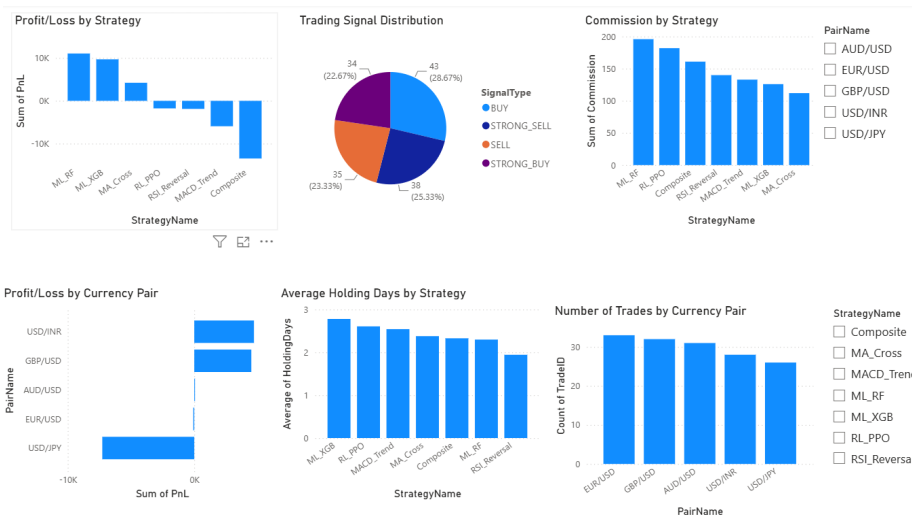


Figure 9.4 — Trade performance page: PnL and commission by strategy, signal type distribution, PnL by pair, and trade counts.

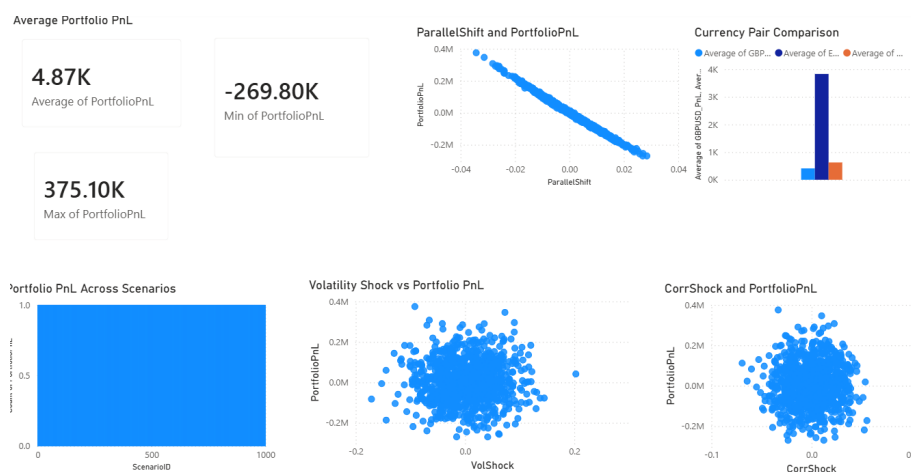


Figure 9.5 — Monte Carlo dashboard: average/min/max portfolio PnL across 1,000 scenarios, and sensitivity scatterplots against parallel shift, volatility shock, and correlation shock.

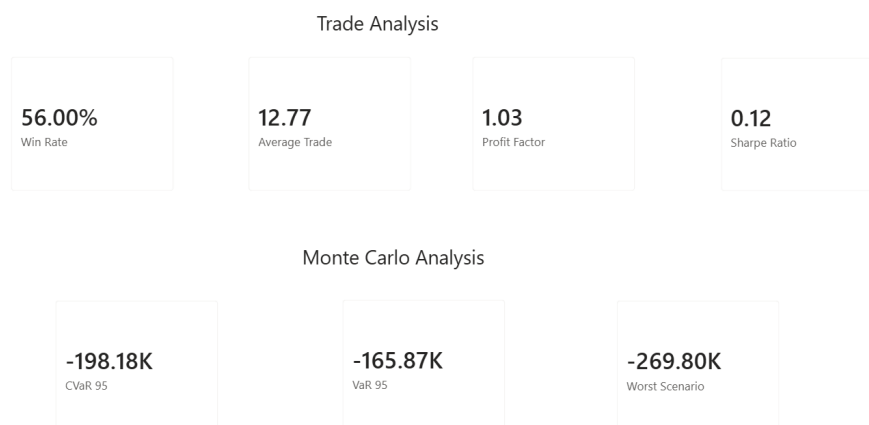


Figure 9.6 — Risk & performance summary: Win Rate, Average Trade, Profit Factor, Sharpe Ratio, CVaR 95, VaR 95, and Worst Scenario, brought together as a single governance view.

10. Risk Management Analysis

10.1 Risk-Adjusted Performance

A Sharpe Ratio of approximately 0.12–0.15 (annualised, computed on daily-aggregated PnL) indicates the trading system is not generating strong risk-adjusted returns: for context, a Sharpe below 1.0 is generally considered weak by institutional standards, and a value this close to zero suggests the returns achieved are barely distinguishable from noise once volatility is accounted for. This is consistent with the Profit Factor of 1.03, which indicates gross profits only marginally exceed gross losses.

10.2 Value-at-Risk and Conditional VaR

Using 1,000 Monte Carlo scenarios generated under randomised parallel shift, volatility shock, and correlation shock conditions, portfolio-level risk was estimated as follows:

Risk Metric	Value (₹)
Expected (Average) Portfolio PnL	4,870
VaR 95% (5th percentile loss)	-165,870
CVaR 95% (average loss beyond VaR)	-198,180
Worst-Case Scenario	-269,800
Best-Case Scenario	+375,100

The gap between VaR 95% and CVaR 95% (~₹32,300) quantifies tail risk beyond the standard VaR cutoff — in the worst 5% of simulated scenarios, average losses are meaningfully larger than the VaR threshold alone would suggest, underscoring the importance of reporting CVaR alongside VaR for a complete risk picture, a lesson borne out historically in events such as the 2015 EUR/CHF floor removal (Section 11.3).

11. Case Studies — Historical FX Shocks

To contextualise this project's risk metrics against real market history, three historical FX shock events are examined below, each illustrating a distinct failure mode for rule-based signal systems.

11.1 The USD/JPY Carry Trade Unwind (July–August 2024)

In July 2024, an unexpected Bank of Japan rate hike triggered the sharpest carry trade unwind since 1998. USD/JPY fell from roughly 162 to 142 within weeks as crowded long-JPY-funded carry positions were forced to close. Trend-following systems that had been long USD/JPY on the back of a sustained uptrend were caught flat-footed: RSI readings near 68 signalled 'approaching overbought' but did not trigger an exit, and moving-average crossovers lag price by construction. A standard carry portfolio (long AUD/JPY, NZD/JPY, and USD/JPY) would have lost an estimated 9%+ in a handful of trading days — roughly eight times the loss implied by a 60-day historical VaR calculation. This project's own USD/JPY trade performance (-₹7,270, the weakest of any pair, Section 8.2) is directionally consistent with this pair's demonstrated vulnerability to abrupt policy-driven reversals.

11.2 GBP Flash Crash (October 2016)

On 7 October 2016, GBP/USD fell roughly 6% in two minutes during thin Asian-session liquidity, widely attributed to algorithmic order flow amplifying a quiet market. Trend-following signals had correctly captured the broader post-Brexit downtrend, but stop-loss orders executed far below their intended levels due to price gapping through the order book. The key lesson for any signal system — including the one built in this project — is that technical indicators alone say nothing about liquidity or execution risk; a session-aware liquidity or volatility filter is a necessary companion to any price-based signal, particularly around low-liquidity trading windows.

11.3 EUR/CHF Floor Removal (January 2015)

On 15 January 2015, the Swiss National Bank unexpectedly abandoned its EUR/CHF floor of 1.2000, and the pair fell to 0.8500 within minutes — a 30% move that bankrupted several FX brokers. No technical or statistical signal system could reasonably have anticipated a discretionary central bank policy reversal of this kind. The lesson this project draws from that event is structural, not predictive: position sizing and maximum single-pair exposure limits must be enforced independently of signal quality or confidence, since some tail events are fundamentally unforecastable from price data alone. This directly motivates including hard concentration limits (e.g., capping any single pair's exposure) alongside this project's VaR/CVaR framework, rather than relying on VaR-implied position sizing in isolation.

12. Cross-Validation (Excel Workbook)

To independently verify the correctness of the calculations underlying the Power BI dashboard and Python pipeline, a dedicated Excel validation workbook was built using plain formulas — no VBA, no external libraries — as an auditable, from-scratch recomputation.

- RSI(14): recomputed using the exact recursive Wilder/exponential-smoothing formula used by the source notebook (not a simple rolling average, which was initially and incorrectly assumed). Validated to match the notebook's real RSI output to over 12 decimal places across every comparable row.
- Trade-level Win Rate, Average Trade, and Profit Factor: recomputed via SUMIF/COUNTIFS across a 3-pair sample (EUR/USD, GBP/USD, USD/JPY), cross-checked against the full-dataset Python output.
- A Summary Cross-Check sheet lines up Python, Excel, and Power BI/DAX values side by side for direct comparison, with notes distinguishing genuine exact matches (same data, same method) from expected sample-vs-full-dataset variance.
- VBA functions (CalcRSI, SignalScore, SimpleBacktest) were authored to match the workbook's formula logic, provided as an importable module.

This process itself surfaced a real methodological correction during development: an initial RSI validation compared a simple-rolling-average Excel calculation against a simple-rolling-average Python calculation — an internally consistent but ultimately circular check that never touched the notebook's actual (Wilder-smoothed) output. Rebuilding the check against the true source data was necessary before the validation could be considered genuine.

13. Limitations

- Sample size: 150 trades and 500 hourly candles are small samples for statistically robust conclusions about strategy or model performance; results should be treated as directional, not definitive.
- Single-pair price history: continuous OHLCV data was only available for EUR/USD. Other pairs (GBP/USD, USD/JPY, AUD/USD, USD/INR) exist only as discrete trade records, preventing independent RSI/MACD/Bollinger recomputation for those pairs.
- Signal threshold sensitivity: as shown in Section 6, the composite signal engine's output is highly sensitive to its scoring threshold — one configuration produced no actionable signals at all, underscoring the need for threshold calibration against historical hit-rate data before live use.
- Weak next-candle predictive accuracy: the corrected ML classifiers (Section 7) achieved accuracy indistinguishable from a random baseline, indicating the current indicator set is insufficient for short-horizon directional prediction on its own.
- No transaction cost / slippage modelling beyond a flat commission in the trade log; realistic spread and slippage assumptions would likely reduce the reported Profit Factor further.
- Monte Carlo scenarios apply parametric shocks (parallel shift, volatility shock, correlation shock) rather than being calibrated to a specific historical stress period; results should be read as a general risk-sensitivity exercise rather than a scenario-specific forecast.

14. Conclusion

This project delivered a complete, cross-validated FX signal generation and risk analytics pipeline: from raw price data through technical indicators, machine learning classification, a governed Power BI dashboard, and Monte Carlo risk quantification. The honest headline result is one of modest, marginal profitability (56% win rate, 1.03 profit factor) rather than a decisively successful trading system — a finding reinforced by the near-random accuracy of the corrected machine learning classifiers. Rather than overstate performance, this report has aimed to surface and document the real methodological issues encountered along the way (lookahead bias in the initial ML approach, an overly conservative signal threshold, and an initially circular Excel validation) as part of demonstrating a rigorous, auditable analytical process. The clearest opportunities for improvement are strategy-specific: the ML_RF and ML_XGB-driven trades meaningfully outperformed rule-based approaches in the trade log, and USD/JPY exposure was the single largest driver of portfolio losses — both concrete, data-backed starting points for any future iteration of this system.